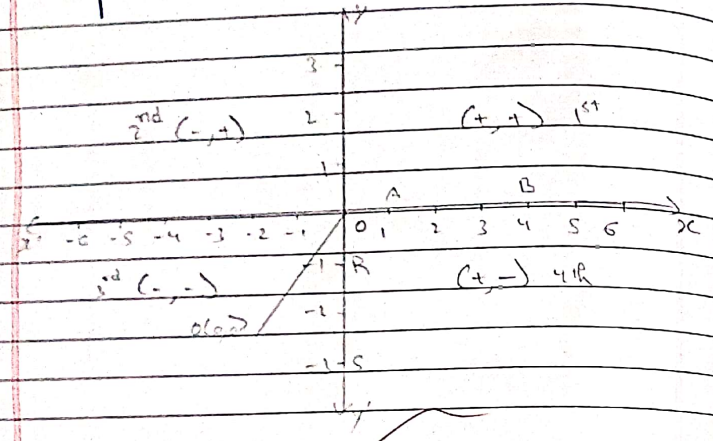
Coordinate Geometry

Recap



(-4,6) → 2nd quad.
(0,3) → on y-axis
(4,0) → on x-axis

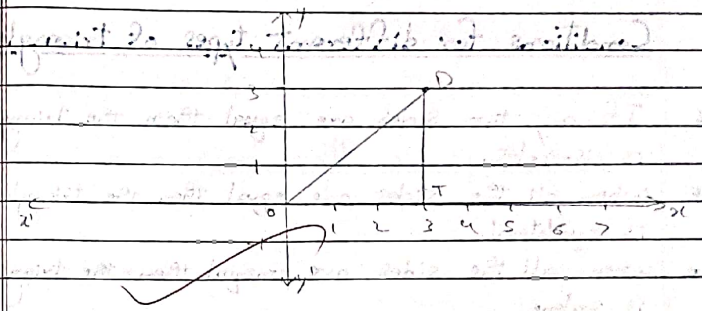
Introduction

* Points are on x-axis
  find distance between AB
  AB = OB - OA
  Answer = 4 - 1 = 3 units

* Points are on Y-axis
  BS = OS - OR = 3 - 1 = 2 units

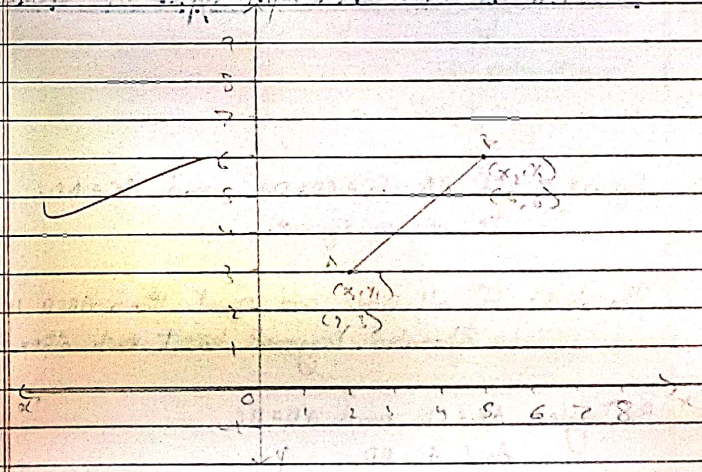
* One point of graph is and other on x & y plain.

△ODT $OD = \sqrt{(OT)^2 + (OT)^2}$
$= \sqrt{(3)^2 + (3)^2}$
$= \sqrt{18} = 3\sqrt{2}$



* Both points on the graph

Distance $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
$= \sqrt{(5-2)^2 + (6-3)^2}$
$= \sqrt{18} = 3\sqrt{2}$



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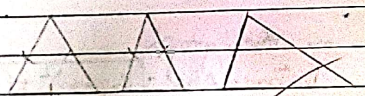
Conditions for Co-linear points -

If AB + BC = AC

A B C

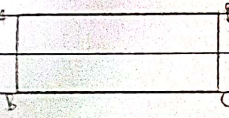
Conditions for different types of Triangle.

* If any two sides are equal then the triangle is isosceles.
* When all the sides are equal then the triangle is equilateral.
* When all the sides are unequal then the triangle is scalene.



Equilateral Isosceles Scalene

Conditions for different types of Quadrilateral

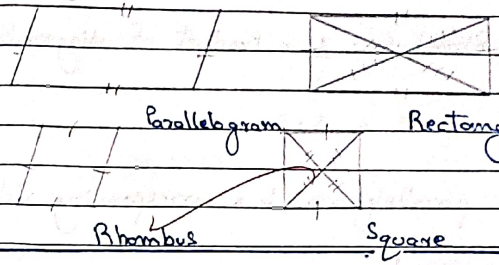


(a) Square - If AB = BC = CD = DA and AC = BD.
It is a square.

(b) Rhombus - If all sides are equal then ABCD is a Rhombus. Diagonals bisect each other at 90°.

(c) Rectangle - AB = CD and AD = BC.
And AC = BD

(d) Parallelogram - AB = CD and AD = BC.

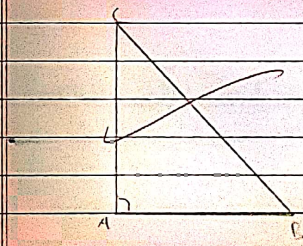


Quadrilateral with all Equal Sides

Square
||
Diagonals are equal

Rhombus

Conditions of Right Triangle



If $BC^2 = AB^2 + AC^2$
the triangle is right Δ.

Area Formulas of Quadrilaterals

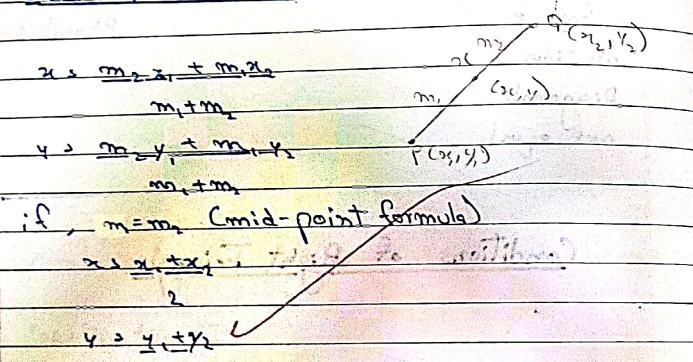
area of Rhombus = $\frac{1}{2}$ × Product of diagonals

area of square = $s \times s$

area of parallelogram = b × corresponding side

area of rectangle = $l \times b$

Section Formula



$$x = \frac{m_2 x_1 + m_1 x_2}{m_1 + m_2}$$

$$y = \frac{m_2 y_1 + m_1 y_2}{m_1 + m_2}$$

if, $m = m_1$ (mid-point formula)

$$x = \frac{x_1 + x_2}{2}$$

$$y = \frac{y_1 + y_2}{2}$$

Concept of Equal division -

(a) Two parts

A ——— B

$(1, 5)$ (x, y) $(-3, 2)$

$$x = \frac{1 + (-3)}{2} = \frac{-2}{2} = -1$$

$$y = \frac{7 + 2}{2} = \frac{9}{2}$$

(b) 3 parts

A P Q B

$(1, 7)$ $(-3, 2)$

(c) 4 parts

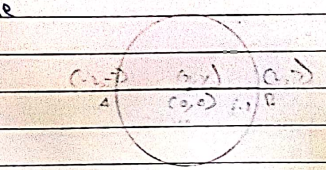
A P Q R B

Process of Finding Center of Circle

Ratio : 1:1

Reason = Radii of Circle

Simplification:



$$0 = \frac{x_1 + x_2}{2}$$

$$0 = \frac{-2 + x_2}{2}$$

$$x_2 = 2$$

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W.S.

Exercise - 7.1

1-i-solu Let the given points be A(2,3) and B(4,1).

∴ Required distance = $AB = \sqrt{(4-2)^2 + (1-3)^2}$

$= \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$ units

ii-Solu Let the given points be A(-5,7) and B(-1,3).

∴ Required distance = $AB = \sqrt{(-1+5)^2 + (3-7)^2}$

$= \sqrt{16+16} = \sqrt{32} = 4\sqrt{2}$ units

iii-solu Let the given points be A(a,b) and B(-a,-b).

∴ Required distance = $AB = \sqrt{(-a-a)^2 + (-b-b)^2}$

$= \sqrt{4a^2 + 4b^2} = 2\sqrt{a^2+b^2}$ units

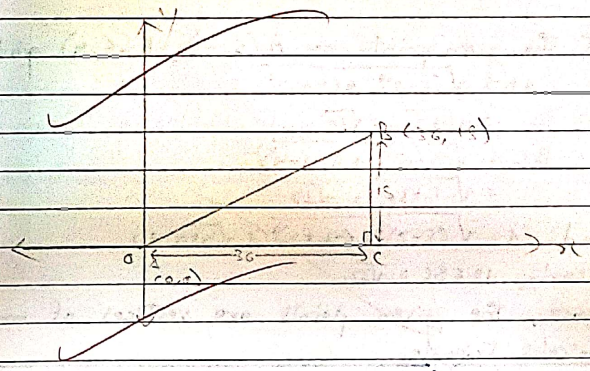
2-solu Let the given points be A(0,0) and B(36,15)

∴ Distance,

$AB = \sqrt{(36-0)^2 + (15-0)^2}$

$= \sqrt{1296+225}$

$= \sqrt{1521} = 39$



Fix the points A(0,0) and B(36,15) as shown in the figure.

Draw BC ⊥ x-axis.

In rt. ∠d △ABC,

$AB = \sqrt{AC^2 + BC^2}$

$= \sqrt{(36)^2 + (15)^2}$

$= \sqrt{1296 + 225}$

$= \sqrt{1521} = 39$

Hence, the required distance between towns A and B = 39 Kms.

3-Solu Let the given points be:

A(1, 5), B(2, 2) and C(−2, −11)

$\therefore AB = \sqrt{(2-1)^2 + (2-5)^2}$

$= \sqrt{1+9} = \sqrt{5}$

$BC = \sqrt{(-2-2)^2 + (-11-2)^2}$

$= \sqrt{16+196} = \sqrt{212}$

and $CA = \sqrt{(1+2)^2 + (5+11)^2}$

$= \sqrt{9+256} = \sqrt{265}$.

Here, the sum of any two is not equal to the third one. Hence, the given points are not collinear.

4-Solu Let the given points be A(5, −2), B(6, 4) and C(7, −2).

$\therefore AB = \sqrt{(6-5)^2 + (4+2)^2}$

$= \sqrt{1+36} = \sqrt{37}$,

$BC = \sqrt{(7-6)^2 + (-2-4)^2}$

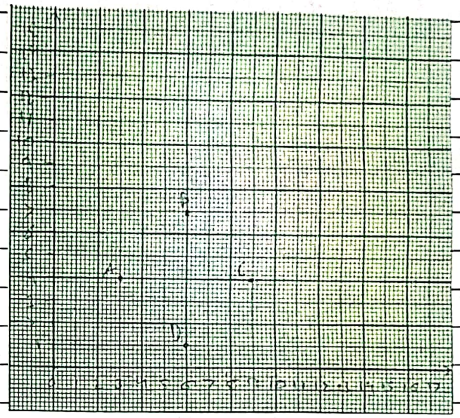
$= \sqrt{1+36} = \sqrt{37}$

and $CA = \sqrt{(5-7)^2 + (-2+2)^2} = \sqrt{4+0} = 2$

Clearly, AB = BC = $\sqrt{37}$.

Hence, the given points are vertices of an isosceles triangle.

5-Solu From the given figure, the given points are A(3, 4), B(6, 7), C(9, 4) and D(6, 1).



$\therefore AB = \sqrt{(6-3)^2 + (7-4)^2} = \sqrt{18}$

$BC = \sqrt{(9-6)^2 + (4-7)^2} = \sqrt{18}$

$CD = \sqrt{(6-9)^2 + (1-4)^2} = \sqrt{18}$

$DA = \sqrt{(3-6)^2 + (4-1)^2} = \sqrt{18}$

$AC = \sqrt{(9-3)^2 + (4-4)^2} = 6$

and $BD = \sqrt{(6-6)^2 + (1-7)^2} = 6$

Thus, AB = BC = CD = DA = $\sqrt{18}$

and AC = BD = 6

Hence, ABCD is a square and as such champs is correct.

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Sol (i) The given points are A(-1,-2); B(1,0); C(-1,2) and D(-3,0).

$\therefore AB = \sqrt{(-1-1)^2 + (-2-0)^2}$

$= \sqrt{(-2)^2 + (-2)^2}$

$= \sqrt{8}$

$BC = \sqrt{(-1-1)^2 + (2-0)^2}$

$= \sqrt{(-2)^2 + (2)^2}$

$= \sqrt{8}$

$CD = \sqrt{(-3+1)^2 + (0-2)^2}$

$= \sqrt{(-2)^2 + (-2)^2}$

$= \sqrt{8}$

$DA = \sqrt{(-1+3)^2 + (-2-0)^2}$

$= \sqrt{4+4} = \sqrt{8}$

$AC = \sqrt{(-1+1)^2 + (2+2)^2}$

$= \sqrt{16} = 4$

and $BD = \sqrt{(-3-1)^2 + (0-0)^2}$

$= \sqrt{16+0} = 4$

Thus, $AB = BC = CD = DA = \sqrt{8}$

and $AC = BD = 4$

Hence, the given quadrilateral is a Square.

(ii) The given points are A(-3,5); B(3,1); C(0,3) and D(-1,-3).

$\therefore AB = \sqrt{(3+3)^2 + (1-5)^2}$

$= \sqrt{36+16} = \sqrt{52} = 2\sqrt{13}$

$BC = \sqrt{(0-3)^2 + (3-1)^2}$

$= \sqrt{(-3)^2 + (2)^2} = \sqrt{13}$

$CA = \sqrt{(0+3)^2 + (3-5)^2}$

$= \sqrt{9+4} = \sqrt{13}$

Thus, $BC + CA = 2\sqrt{13} = AB$

$\therefore$ A, B and C are collinear.

Hence, the given points do not form any quadrilateral.

(iii) The given points are A(4,5); B(7,5); C(4,1) and D(1,2).

$\therefore AB = \sqrt{(7-4)^2 + (5-5)^2}$

$= \sqrt{9+1} = \sqrt{10}$

$BC = \sqrt{(4-7)^2 + (1-5)^2}$

$= \sqrt{(-3)^2 + (-4)^2} = \sqrt{18} = 3\sqrt{2}$

$CD = \sqrt{(1-4)^2 + (2-1)^2}$

$= \sqrt{(-3)^2 + (1)^2} = \sqrt{10}$

$DA = \sqrt{(4-1)^2 + (5-2)^2}$

$= \sqrt{(3)^2 + (3)^2} = 3\sqrt{2}$

$AC = \sqrt{(4-4)^2 + (1-5)^2}$

$= \sqrt{(-4)^2} = 4$

and $BD = \sqrt{(1-7)^2 + (2-5)^2}$

$= \sqrt{(-6)^2 + (-3)^2} = 3\sqrt{13}$

Thus, $AB = CD$ and $BC = DA$ but $AC \neq BD$

i.e., opposite sides are equal but the diagonals are not equal.

Hence, the given quadrilateral is a parallelogram.

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7-sol) Let P(x, 0) be any point on the x-axis.
Now P(x, 0) is equidistant from the points A(2, -5) and B(-2, 9).

$\therefore$ AP = BP

$\Rightarrow \sqrt{(x-2)^2+(0+5)^2} = \sqrt{(x+2)^2+(0-9)^2}$

$\Rightarrow (x-2)^2+25 = (x+2)^2+81$ (Squaring)

$\Rightarrow x^2+4-4x+25 = x^2+4+4x+81$

$\Rightarrow -8x = 81-29$

$\Rightarrow x = \frac{52}{-8}$

$\Rightarrow x = -7$

Hence, the required points is (-7, 0).

8-sol) The given points are P(2, -3) and Q(10, y).

$\therefore PQ = \sqrt{(10-2)^2+(y+3)^2}$

$= \sqrt{64+y^2+9+6y}$

$= \sqrt{y^2+6y+73}$

By the question

$PQ = 10$

$\Rightarrow \sqrt{y^2+6y+73} = 10$

$\Rightarrow y^2+6y+73 = 100$ (Squaring)

$\Rightarrow y^2+6y-27 = 0$

$\Rightarrow y^2+9y-3y-27 = 0$

$\Rightarrow y(y+9)-3(y+9) = 0$

$\Rightarrow (y-3)(y+9) = 0$

$\Rightarrow y = 3$ or -9

Hence, $y = -9, 3$.

9-sol) The given points are Q(0, 1), P(5, 3) and R(x, 6).

$\therefore QP = \sqrt{(5-0)^2+(3-1)^2}$

$= \sqrt{25+16} = \sqrt{41}$

and $QR = \sqrt{(x-0)^2+(6-1)^2}$

$= \sqrt{x^2+25}$

By the Question,

$QP = QR$

$\Rightarrow \sqrt{41} = \sqrt{x^2+25}$

$\Rightarrow 41 = x^2+25$ (Squaring)

$\Rightarrow x^2 = 16$

$\Rightarrow x = \pm 4$

(I) when $x = 4$, then R is (4, 6).

$\therefore QR = \sqrt{(4-0)^2+(6-1)^2}$

$= \sqrt{16+25} = \sqrt{41}$

and $PR = \sqrt{(4-5)^2+(6+3)^2}$

$= \sqrt{1+81} = \sqrt{82}$

(II) when $x = -4$, then R is (-4, 6).

$\therefore QR = \sqrt{(-4-0)^2+(6-1)^2}$

$= \sqrt{16+25} = \sqrt{41}$

and $PR = \sqrt{(-4-5)^2+(6+3)^2}$

$= \sqrt{81+81} = 9\sqrt{2}$

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10-Solu The given points are A(3, 6) and B(-3, 4).
Let P(x, y) be the points such that PA = PB ---①
Now $PA = \sqrt{(x-3)^2 + (y-6)^2}$
and $PB = \sqrt{(x+3)^2 + (y-4)^2}$
Putting in ①, $\sqrt{(x-3)^2 + (y-6)^2} = \sqrt{(x+3)^2 + (y-4)^2}$
Squaring, $(x-3)^2 + (y-6)^2 = (x+3)^2 + (y-4)^2$
$\Rightarrow (x^2-6x+9) + (y^2-12y+36) = (x^2+6x+9) + (y^2-8y+16)$
$\Rightarrow -6x-12y+45 = 6x-8y+25$
$\Rightarrow -12x-4y+20 = 0$ ---②
$\Rightarrow 3x+y-5 = 0$
which is the required relation.

Exercise 7.2

1-Solu Let P(x, y) be the point, which divides the join of given points A(-1, 7) and B(4, -3) in the ratio 2:3.

$\therefore x = \frac{2(4)+3(-1)}{2+3} = \frac{8-3}{5}$

$= 1$

$y = \frac{2(-3)+3(7)}{2+3} = \frac{-6+21}{5}$

$= 3$

Hence, the required point is (1, 3)

2-Solu Let P and Q be the points of trisection of AB i.e. AP = PQ = QB.
$\therefore$ P divides line segments AB in the ratio 1:2 internally.
$\therefore$ Coordinates of P are:

$\left[\frac{1(-2)+2(4)}{1+2}, \frac{1(-3)+2(-1)}{1+2}\right]$

i.e., $\left[\frac{-2+8}{3}, \frac{-3-2}{3}\right]$

i.e., $\left[2, -\frac{5}{3}\right]$

And Q divides line segment AB in the ratio 2:1 internally.
$\therefore$ Coordinates of Q are:

$\left[\frac{2(-3)+1(4)}{2+1}, \frac{2(-3)+1(-1)}{2+1}\right]$

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i.e., $\left[\frac{-4+4}{3}, \frac{-6-1}{3}\right]$ i.e., $\left[0, \frac{-7}{3}\right]$

Hence, the coordinates of the points of trisection of the line segment joining A and B are $\left[2, \frac{-5}{3}\right]$ and $\left[0, \frac{-7}{3}\right]$.

3-Solu In the given figure, we take A as origin. Taking x-axis along AB and y-axis along AD.

Position of green flag

= Distance covered by Niharika

= Niharika runs $\frac{1}{4}$th distance AD on the 2nd line.

$= \frac{1}{4} \times 100 = 28$ m

∴ Coordinates of the green flag are (2, 28).

Position of red flag = Distance covered by Preet

= Preet runs $\frac{1}{5}$th distance AD on the 8th line

$= \frac{1}{5} \times 100 = 20$ m

∴ Coordinates of the red flag are (8, 20)

∴ Distance between Green and Red flags

$= \sqrt{(8-2)^2 + (20-28)^2}$

$= \sqrt{36+25} = \sqrt{61}$ m

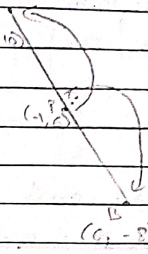
Position of blue flag

= Mid-point of green flag and red flag

$= \left[\frac{2+8}{2}, \frac{28+20}{2}\right] = (5, 22.5)$.

Hence, blue flag is in the 5th line and at a distance of 22.5 m along AD.

4-Solu Let P(-1, 6) divide the line segment joining the points A(-3, 10) and B(6, -8) in the ratio k:1.



$\therefore \frac{6k+1(-3)}{k+1} = -1$

$\Rightarrow 6k - 3 = -k - 1$

$\Rightarrow 7k - 2 = 0$

$\Rightarrow k = \frac{2}{7}$

Hence, the required ratio is 2:7.

5-Solu Let the ratio be k:1.

By section Formula, the coordinates of the point, which divides AB in the ratio k:1 are:

$\left[\frac{k(-4)+1(1)}{k+1}, \frac{k(5)+1(-5)}{k+1}\right]$

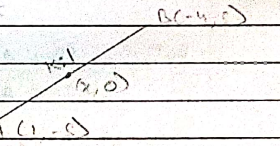
$= \left[\frac{-4k+1}{k+1}, \frac{5k-5}{k+1}\right]$

This point lies on x-axis (i.e., $y = 0$),

$\therefore \frac{5k-5}{k+1} = 0$

$5k = 5$

$k = 1$



Hence, the required ratio ratio is 1:1 and the of intersection is:

$\left[\frac{1-4}{1+1}, \frac{-5+5}{1+1}\right]$ i.e., $\left[\frac{-3}{2}, 0\right]$

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6-Solu Let A(1,2), B(4,y), C(x,6) and D(3,5) be the vertices of the parallelogram ABCD.

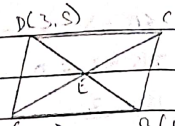
⇒ Mid-point of AC = Mid-point of BD

$\Rightarrow \left(\frac{1+x}{2}, \frac{2+6}{2}\right) = \left(\frac{4+3}{2}, \frac{y+5}{2}\right)$

$\Rightarrow \frac{1+x}{2} = \frac{7}{2}$ and $\frac{8}{2} = \frac{y+5}{2}$

$\Rightarrow 1+x = 7$ and $8 = y+5$

Hence, $x = 6$ and $y = 3$



7-Solu Let the coordinates of A be (x, y).

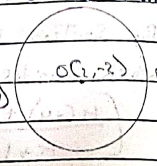
Since the centre O is the mid-point of the diameter AB.

$\therefore \frac{x+1}{2} = 2$ and $\frac{y+4}{2} = -3$

$\Rightarrow x+1 = 4$ and $y+4 = -6$

$\Rightarrow x = 3$ and $y = -10$

Hence, the coordinates of A are $(3, -10)$.



8-Solu Let (x, y) be the coordinates of P.

Given: $AP = \frac{3}{7} AB$.

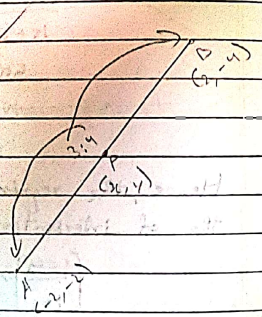
But, $PB = AB - AP$

$= AB - \frac{3}{7} AB$

$= \left(1 - \frac{3}{7}\right) AB$

$= \frac{4}{7} AB$.

$\frac{AP}{PB} = \frac{3/7\ AB}{4/7\ AB} = \frac{3}{4}$



Thus, P divides the join of A and B in ratio 3:4

$\therefore x = \frac{3(4) + 4(-2)}{3+4}$

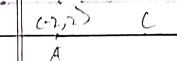
and $y = \frac{3(-4) + 4(-2)}{3+4}$

$\Rightarrow x = \frac{6-8}{7}$ and $y = \frac{12-8}{7}$

$\Rightarrow x = \frac{-2}{7}$ and $y = \frac{-20}{7}$

Hence, the coordinates of P are $\left(\frac{-2}{7}, \frac{-20}{7}\right)$.

9-Solu Let C, D and E be the points, which divide the line segment joining A(-2, 2) and B(2, 8) into four equal parts.



Then D is the mid-point of AB,

C is the mid-point of AD,

and E is the mid-point of DB,

where AC = CD = DE = EB.

Since D is the mid-point of AB,

$\therefore$ Coordinates of D are $\left(\frac{-2+2}{2}, \frac{2+8}{2}\right)$

i.e., $(0, 5)$

Since C is the mid point of AD,

$\therefore$ Coordinates of C are:

$\left(\frac{-2+0}{2}, \frac{2+5}{2}\right)$ i.e., $\left(-1, \frac{7}{2}\right)$

Since E is the mid-point of DB,

$\therefore$ coordinates of E are $\left(\frac{0+2}{2}, \frac{5+8}{2}\right)$ i.e., $\left(1, \frac{13}{2}\right)$

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Hence, the required points are: $C\left(-1, \frac{7}{2}\right)$, $O(0,0)$ and $P\left(1, \frac{15}{2}\right)$.

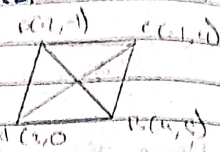
10. Sol. Let A(2,0), B(4,8), C(-1,4) and D(-2,-1) be the vertices of the rhombus ABCD.

$\therefore AC = \sqrt{(-1-2)^2 + (4-0)^2}$

$= \sqrt{16+16} = 4\sqrt{2}$

and $BD = \sqrt{(-2-4)^2 + (-1-8)^2}$

$= \sqrt{36+36} = 6\sqrt{2}$



$\therefore$ Area of rhombus ABCD $= \frac{1}{2} \times AC \times BD$

$= \frac{1}{2}(4\sqrt{2} \times 6\sqrt{2}) = \frac{1}{2}(48) = 24$

Hence, the area of rhombus ABCD = 24 sq. units.

W.S.